

CRITICAL REVIEW

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A comprehensive review of Dynamic Chemical Labelling on Luminex xMAP technology: a journey towards Drug-Induced Liver Injury testing

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Drug-Induced Liver Injury (DILI) is a grave global adverse event that can result in fatal consequences, causing drug failures, market withdrawals, and regulatory warnings, leading to substantial financial losses. The early detection of DILI remains a significant challenge in global healthcare. Although circulating microRNAs (miRs) show promise as clinical biomarkers for DILI, the current analytical methods for their measurement are insufficient. There is a pressing need for rapid and reliable miR detection methods that eliminate the need for nucleic acid extraction and PCR-based amplification. This review highlights recent advancements achieved by integrating Dynamic Chemical Labelling (DCL) with Luminex xMAP technology. This powerful combination has resulted in groundbreaking bead-based assays that allow (1) the direct, multiplex detection of miRs, and (2) the simultaneous testing of miR and protein biomarkers. This triple capability enables a comprehensive assessment that significantly enhances the detection and analysis of crucial biomarkers, thus improving the understanding and diagnosis of DILI. In conclusion, this review offers valuable insights into the capabilities and potential applications of these groundbreaking assays in DILI research, as well as their potential use in other diagnostic and research domains that require direct or multiplex analysis of miRs or analysis of miRs in combination with proteins.

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Introduction

Adverse drug reactions (ADR) are a significant concern for patients, healthcare professionals, and the pharmaceutical industry, with an estimated annual cost of €79 billion in the EU.¹ Drug-induced liver injury (DILI) is a prevalent and severe ADR and is a leading cause of acute liver failure (ALF) in the Western World.^{2–5} ALF is a life-threatening condition, and identifying the risk of developing ALF is a top priority since it is frequently associated with high morbidity and mortality rates.⁶

The incidence of DILI depends on various factors, such as the drug itself and patient-specific factors like sex, ethnicity, or genetic polymorphism in drug detoxification.⁷ Overdose of paracetamol, also known as acetaminophen, is the leading cause of DILI in developed countries.⁸ Patients with paracetamol-induced liver injury often report to accident and emergency centers at hospitals and require admission to intensive care, which incurs a considerable cost.^{9,10}

Due to its potent antioxidant properties, particularly in enhancing endogenous glutathione (GSH) content to counteract oxidative stress, *N*-Acetyl Cysteine (NAC) has gained significant attention as a therapeutic agent against DILI.^{11–14}

However, treatment with NAC takes 20 to 25 hours to complete and can produce a high incidence of adverse reactions.¹⁵ If paracetamol-induced injury could be ruled out early, unnecessary, and potentially harmful treatment could be avoided, and hospital bed availability could be substantially increased.

Also, amoxicillin-clavulanate (AC) has been linked to hundreds of cases of clinically apparent acute liver injury and is currently the most common cause of DILI in most large case series in the United States and Europe.¹⁶ The antibiotic amoxicillin-clavulanate results in approximately 1 in 2300 patients developing DILI.¹⁷ The US Drug Induced Liver Injury Network (DILIN)¹⁸ and the Spanish DILI Registry¹⁹ have identified AC as the most common cause of non-acetaminophen DILI.^{18,20}

Furthermore, DILI is a significant contributor to drug attrition across all stages of the drug discovery process.²¹ Notably, in early development, up to 50% of all preclinical candidate drugs display hepatotoxic effects at supra-therapeutic doses.⁷ In a decade, DILI has been the cause of 18% of all post-marketing drug withdrawals, rendering it the principal reason for such actions.²² In the EU and USA between 1997 and 2016, eight drugs were withdrawn due to DILI-associated incidents, resulting in liver transplants and fatalities.²² The identification of suspected hepatotoxicity in research and clinical trials is challenging, as heightened levels of hepatic enzymes do not always stem from DILI, but from other underlying liver pathologies.²³

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The diagnosis of DILI currently relies on expert opinion, which considers patient data and the characteristic “signatures” associated with certain drugs.²⁴ Causality scores, such as the Roussel Uclaf Causality Assessment Method (RUCAM), are designed to confirm or exclude the suspicion of DILI.²⁵ However, RUCAM has several limitations, including its complex scale (six sections, each with its own point system), time-consuming completion, ambiguous definitions, and lack of data to support the selection and weighting of component domains.⁸ Furthermore, RUCAM’s reliance on drug re-challenge data and the potential for liver test fluctuations in patients with underlying liver disease or taking multiple drugs can lead to false-negative results.⁹ Other causality scores, such as Hy’s law and the Model for End-Stage Liver Disease (MELD) score, are also used.^{26–28} Positive cases identified by Hy’s law are of great concern due to the risk of mortality, while the MELD score prioritizes individuals for liver transplantation. Nonetheless, the MELD score does not use transaminases ALT/AST to define individuals in urgent need of transplantation.²⁸ Most drug makers agree that the common liver function tests, including transaminases and other serum biochemistries, are poor predictors of a drug’s potential to cause ALF.²⁹

Given the limitations associated with causality scores, there is an urgent and unmet need for new and improved diagnostic biomarkers in the form of blood tests to help researchers, regulators, and the pharmaceutical industry assess the risk of DILI and its potential to cause liver failure reliably.³⁰ Consequently, several industry-wide initiatives have been aimed at finding and validating improved biomarkers for DILI, including the European Medicines Agency Safe-T consortium, TransBio-line, and the Pro-Euro DILI NETWORK.^{31,32} These initiatives aim to predict and diagnose drug-induced hepatotoxicity before market authorization and support DILI’s post-authorization monitoring and clinical management. However, a recent letter of support from the EMA has highlighted the ongoing lack of sensitive and specific assays to diagnose, predict, and monitor idiosyncratic DILI as a severe hurdle in drug development.³³

A success story in the development of new biomarkers for DILI is miR-122-5p, which constitutes a substantial proportion of all miR transcripts in the liver and is specific to this tissue.³⁴ Research has demonstrated that miR-122-5p is elevated in the plasma of patients with established paracetamol-induced liver injury and can identify liver injury at an early time when current markers such as ALT are still normal.³⁵ Compared to conventional hepatotoxic markers, circulating miR-122-5p can effectively distinguish intrahepatic from extrahepatic damage with higher sensitivity and specificity.³⁶ miR-122-5p is expected to be a valuable pre-clinical and clinical biomarker of DILI.³⁴ Emerging scientific research suggests that the integration of miR-122-5p with both traditional and contemporary protein markers can markedly enhance the early detection of liver injury and assist in determining its origin.

The absence of universally recognized diagnostic tests makes DILI diagnosis a persistent challenge in both clinical practice and pharmacovigilance. As a result, there is a growing recognition of the potential of advanced systems in detecting and analysing miR-122-5p for DILI diagnosis. In 2022, Kersaudy-

Kerhoas and team introduced a cartridge-based system employing electrochemical impedance spectroscopy to detect miR-122-5p in DILI patients ref. 37. This innovative platform streamlines the sample preparation process and enables miR detection promptly after a blood draw, proving beneficial even in limited-resource settings. Nonetheless, it is worth noting that this approach necessitates initial extraction and amplification stages, bringing along the intrinsic challenges of PCR technology.

In a separate 2022 study, X. Li and colleagues demonstrated a groundbreaking method for the highly sensitive and label-free detection of the DILI biomarker miR-122-5p, leveraging target recycling and primer exchange reaction (PER) amplifications.³⁸

Furthermore, in 2020, Wang *et al.* delved into a novel method that studies the combination of protein and miRNA spike-ins.³⁹ However, the concurrent detection of both these entities from actual clinical samples has yet to be documented. Notably, their method was validated using only spike-in miRNAs and proteins within serum samples.³⁹

In this review, a groundbreaking advancement in the field of DILI testing is presented, achieved through the powerful combination of Dynamic Chemical Labelling (DCL) and Luminex xMAP technology. This innovative approach results in the development of bead-based tools capable of specifically identifying DILI-specific miR and protein biomarkers, addressing the critical need for earlier and more sensitive DILI analysis.

The combination of DCL and Luminex xMAP technology has created unique tools for DILI analysis. Firstly, a streamlined and efficient single-plex assay has been successfully developed, enabling the direct measurement of circulating miR-122-5p without the requirement for nucleic acid extraction and amplification (Fig. 1A). Secondly, the scope of biomarkers that can be simultaneously analysed without nucleic acid extraction and amplification has been expanded through the creation of a multi-plex assay, allowing for the measurement of miR-122-5p along with other miRs such as miR-451a-5p and miR-193a-5p (Fig. 1B). Lastly, a unique assay called “seqCOMBO” has been introduced, facilitating the simultaneous detection of miR-122-5p and protein arginase 1 (ARG1), providing a comprehensive assessment of DILI biomarkers (Fig. 1C).

The review summarises the development, performance, and validation of these groundbreaking tools using clinical samples. The potential of the Luminex platform for early detection of DILI is emphasised, underlining the importance of timely intervention and improved patient outcomes. Furthermore, the findings pave the way for the future implementation of simplified, cost-effective, and robust multi-plex tests capable of detecting multiple miRs or combinations of miR and protein biomarkers.

In conclusion, this review showcases remarkable advancements in DILI testing achieved through the integration of DCL with Luminex xMAP technology. The presented tools offer a transformative approach to DILI analysis, providing valuable insights into their capabilities and potential applications in DILI research and other diagnostic and research areas that require direct or multi-plex analysis of miRs or the analysis of miRs in combination with proteins.

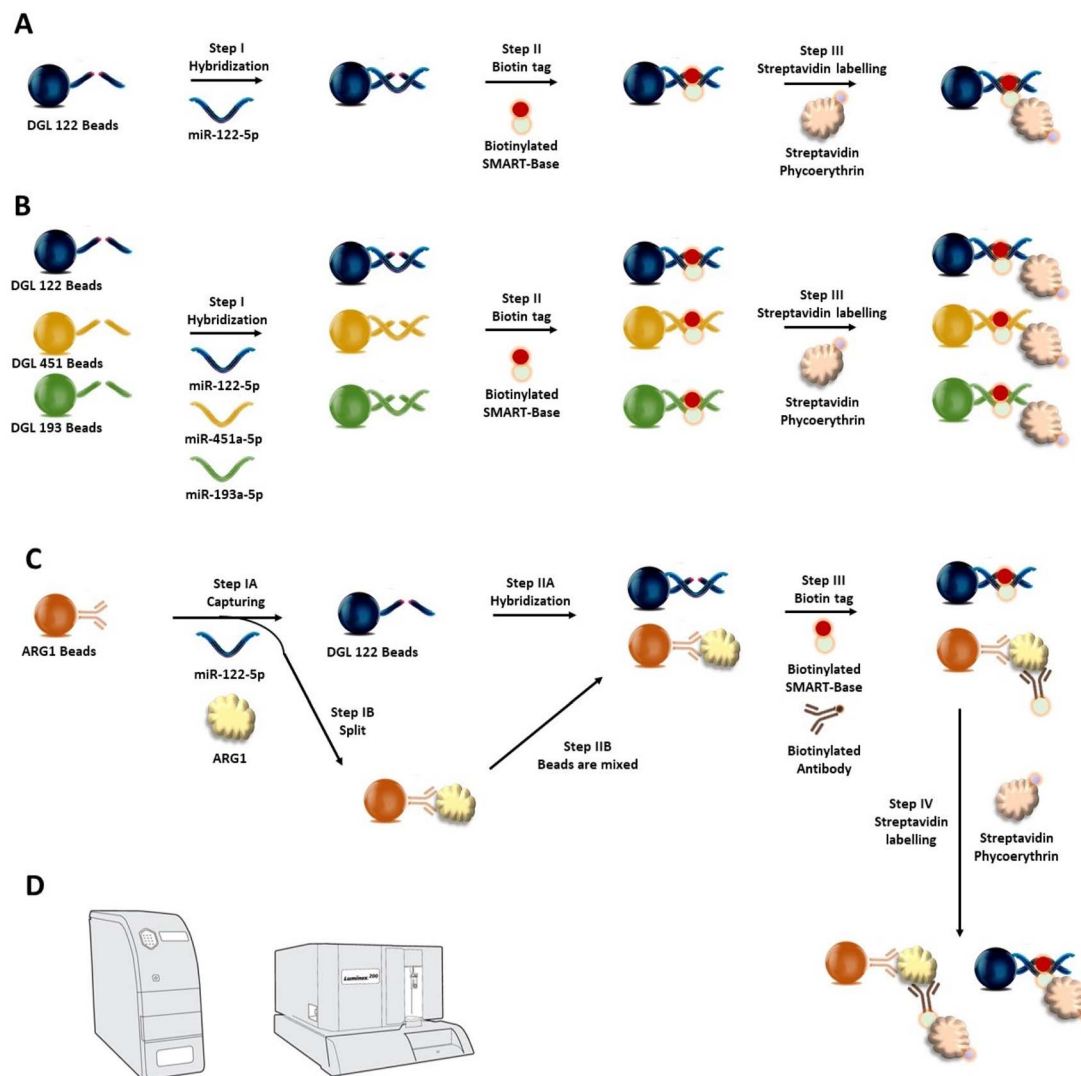


Fig. 1 Cartoon representing general workflows of the three novel tools. (A) Direct measurement of circulating miR-122-5p in a single-plex format: (Step I) DGL 122 beads hybridization with complementary miR-122-5p targets; (Step II) DCL reaction with SMART-C biotin incorporation; (Step III) DGL 122 beads labelling with SA-PE. Total workflow time 3.5 hours. (B) Analysis of miR-122-5p, miR-451a-5p and miR-193a-5p in the multi-plex format: (Step I) DGL 122, 451 and 193 beads hybridization with complementary miR-122-5p, miR-451a-5p and miR-193a-5p targets; (Step II) DCL reactions with SMART-C biotin incorporation; (Step III) DGL 122, 451 and 193 beads labelling with SA-PE. Total workflow time 3.5 hours. (C) Detection of miR-122-5p in combination with protein arginase 1 (seqCOMBO). Step (IA) — ARG1 beads are added to the serum sample; step (IB) — ARG1 beads capture the ARG1 and separated from the supernatant; step (IIA) — the supernatant is added with Stabiltech buffer containing DGL 122 beads that hybridize with complementary miR-122-5p targets; step (IIB) — both sets of beads are merged; step (III) — the detection antibody and SMART-C biotin are added to the mixture of ARG1 beads and DGL 122 beads for the biotin labelling; step (IV) — ARG1 beads and DGL 122 beads are labelled with Streptavidin–Phycoerythrin (SA-PE). Total workflow time 4.5 hours. For the three tools (A–C), beads are finally read-out by measuring values of MFI using the Luminex MAGPIX system (D).

Dynamic Chemical Labelling

Our research group has developed the unique DCL method for directly detecting and quantifying single-stranded nucleic acids (ssNAs) in bodily fluids, particularly circulating miRs. DCL utilizes a combination of biotinylated aldehyde-modified nucleobases (SMART-Bases) and modified PNA capture probes synthesized with an abasic position (DGL-Probes) to interrogate complementary ssNAs by applying a dynamic chemistry reaction, as illustrated in Fig. 2.^{40–42} This chemical method offers single base resolution accuracy.^{36,43,44} One of the major

advantages of DCL is its ability to minimize or eliminate false positives by requiring two specific molecular events to generate a signal: perfect hybridization between ssNAs and the DGL-Probe, and specific molecular recognition by the SMART-Base, based on Watson–Crick base-pairing rules (Fig. 2).^{40–54}

Over the years, the DCL method has been combined with various bead-based platforms to analyze the expression of circulating miRs in biological fluids (Table 1).

Particularly, the DGL method has been merged with Luminex xMAP technology to enable the detection of miRs in combination with proteins and allows for both single- and

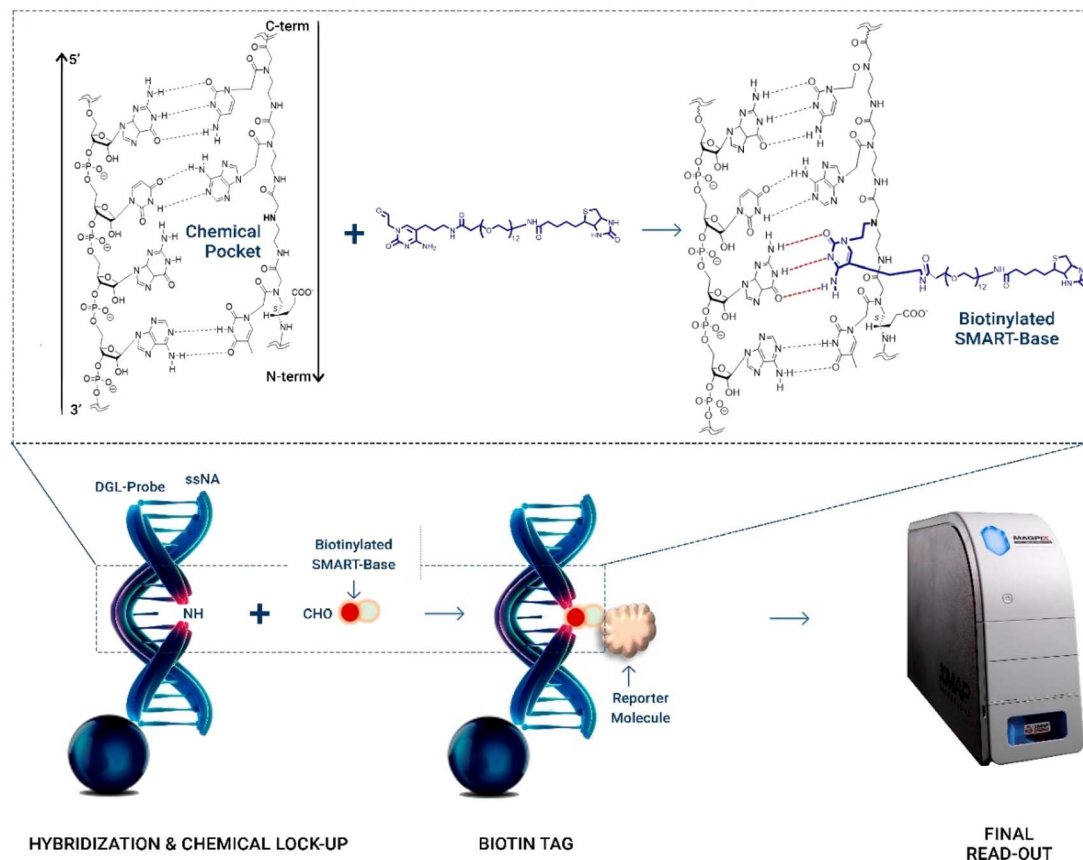


Fig. 2 Cartoon illustrating the general workflow of Dynamic Chemical Labelling (DCL). The DGL-Probe captures the complementary single-stranded nucleic acid (ssNA) sequence (represented by miR in this review), forming the Chemical Pocket. Once the target ssNA is fully hybridized, the biotinylated SMART-Base is incorporated and covalently linked to the backbone of the DGL-Probe, resulting in the chemical lock-up. The duplex is then detected using a reporter molecule, such as streptavidin–phycoerythrin in the case of the Luminex platform, which specifically recognizes the biotin tag. The final read-out of the DCL reaction is obtained using the Luminex MAGPIX. The top part of this figure provides a magnified view of the cartoon, showcasing the chemical structure of the biotinylated SMART-Base that is incorporated into the Chemical Pocket.

multi-plex analysis. These advanced methods will be thoroughly reviewed in this article.

Luminex xMAP technology

xMAP technology is a versatile and high-throughput platform that allows for the simultaneous detection of multiple analytes within a single complex sample.^{56–58} It currently supports various nucleic acid and immunoassay analyses formats, enabling the detection and typing of pathogenic viruses, bacteria, parasites, fungi, antigens, and antibodies.^{57–64} As an open-architecture platform, xMAP technology is widely used in pharmaceutical, clinical, and research laboratories, providing end-users with flexibility and cost-effectiveness.^{65,66} The xMAP technology employs fluorescent-coded microspheres, where each capture molecule is covalently linked to a unique color-coded microsphere.^{67,68} The reading device can distinguish each bead set based on its color and classify each set separately. Using flow cytometric technology, the Luminex instrument typically measures the laser-induced fluorescent signals emitted from each bead.^{69,70} But the Luminex MAGPIX system offers an

alternative method that uses color-coded magnetic microspheres displayed in a monolayer. The MAGPIX system detects the microspheres using a light-emitting diode (LED) and captures their images using a charge-coupled device (CCD) camera (Fig. 3).⁶⁴ This technology significantly reduces the cost of multi-plex assays, as the analyte is immobilized on a small surface area to be analysed, and sample volumes are minimized compared to ELISA.⁵⁸

For nucleic acid testing, Luminex has developed the MagPlex-TAG color-coded beads, which are fluorescent-labelled magnetic microspheres pre-coupled with xTAG oligonucleotide sequences to interrogate ssNAs.⁶⁹

miR analysis using the Luminex MagPlex-TAG color-coded microspheres relies on three steps shown in Fig. 4: (A and B) the biotin-labelled chimeric probe with an anti-TAG sequence hybridise to the complementary mature miR target (step-down probe hybridisation); (C) MagPlex-TAG Microsphere with anti-TAG sequence hybridises with its 100% complementary TAG sequence on the chimeric probe with the mature miR target (microsphere hybridisation); (D) MagPlex-TAG Microsphere is

Table 1 Integration of Bead-based platforms with the DCL method for miR analysis

Platform	Type of detection	miR target	Type of sample	Sample matrix volume (μL)	Limit of detection (fmols)	Time to results (hours)	Reference
Luminex MAGPIX	Fluorescence	miR-122, miR-451a and miR-193a	Serum	10–25	0.05–0.20	3.5	43, 51, 52 and 54
Quanterix SR-X	Fluorescence	miR-122	Serum, plasma, cell culture supernatant	10	0.05–0.20	3.5	36 and 50
SMCXP	Fluorescence	miR-122	Serum	10–25	0.05–0.20	3.5	55
FLUOstar omega	Fluorescence	miR-451a	Serum, plasma, cell culture supernatant	5	3.5–7.0	3.5	45 and 46
ODG platform	Chemiluminescence	miR-21	Serum, plasma	100	0.1	3.5	47
Flow cytometry (FACS)	Fluorescence	miR-21	Cell culture supernatant	100	n.d.	3.5	46

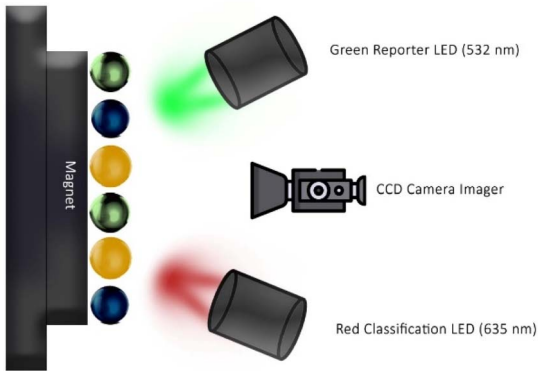


Fig. 3 Principle of analysis by the Luminex MAGPIX fluorescent imager. Color-coded magnetic microspheres displayed in a mono-layer are irradiated by LEDs, and a CCD camera records the picture (LED: light-emitting diode; CCD: charge-coupled device).

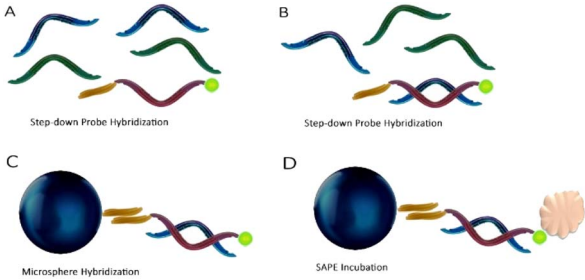


Fig. 4 miR assay workflow via MagPlex-TAG Microspheres. (A and B) Step-down probe hybridization – chimeric probe, with an anti-TAG sequence and biotin labelled, hybridize to target miR during incremental reductions in annealing temperature; (C) microsphere capture – miR-chimeric probe complex is hybridized to microspheres; (D) Steptavidin–Pycoerythrin (SA–PE) incubation – a brief incubation with SA–PE incorporates reporter molecules and detection – targets of interest are quantified on an xMAP instrument.

incubated with Steptavidin–Pycoerythrin (SA–PE) for the final read-out by Luminex instrument.

While Luminex MagPlex-TAG Microspheres are a valuable technology for testing DNA PCR products,^{71,72} researchers have encountered challenges when analysing short RNAs, particularly miRs. This can present significant obstacles for those requiring high specificity and resolution in their experiments. The method relies on hybridization, which can lead to errors in detecting closely related miR targets that differ by only a single base.⁵⁰ In addition, conventional Luminex protocol for miR analysis includes extensive pre-processing of samples. This pre-processing includes steps like extraction, reverse transcription, and amplification, which collectively contribute to a time-consuming process. Typically, Luminex protocols necessitate around 6 hours from sample processing to obtaining results. This prolonged duration is time-consuming and heightens the risk of sample degradation or loss. In the context of DILL, where timely information is of utmost importance for clinical decision-making, the need for a more efficient and rapid methodology becomes imperative.

Therefore, our group has extensively utilized the DCL method on the Luminex platforms, specifically developed to overcome the limitations of short RNA analysis, especially for DILI miR and protein biomarkers, as previously discussed. Thanks to the innovation provided by the DCL on the Luminex platform, it is now possible to detect miRs rapidly and accurately in both single- and multi-plex assays, as well as in combination with proteins, with high sensitivity, without the need for extraction or amplification.

In the following sections, we will describe three innovative tools that leverage the advantages of this amplification-free and DCL chemical nucleic acid testing on the Luminex MAGPIX platform.

Combination of Dynamic Chemical Labelling (DCL) and Luminex xMAP technology

This review showcases the synergistic integration of DCL technology with the Luminex MAGPIX platform for miR analysis, combining their individual strengths. The DCL method provides an amplification-free approach with exceptional specificity and single-base resolution. At the same time, the Luminex xMAP technology enables the simultaneous detection of multiple miRs and facilitates protein analysis simultaneously. The review specifically focuses on the development of three innovative tools utilising this combined technology for the analysis of miR-122-5p in single-plex, miR-122-5p, miR-451a-5p and miR-193a-5p in multi-plex, and the simultaneous measurement of protein arginase 1 (ARG1) and miR-122-5p (seqCOMBO), all validated using clinical samples.

Testing miR-122-5p in single-plex

In 2016, an article was published⁴³ describing the development of the first single-plex analysis of miR-122-5p on Luminex MAGPIX. This was achieved by merging the DCL with Luminex xMAP technology. The successful conjugation of the DGL-Probes to MagPlex microspheres and subsequent direct detection of spike-in synthetic mimic miR-122-5p oligomer *via* DCL demonstrated the potential for rapid and accurate profiling of not only miR-122-5p but also miRs in general. With a Limit of Detection (LoD) of 15 fmoles and the capacity to differentiate sequences containing a single nucleobase mismatch. This Proof-of-Concept (PoC) allowed for direct detection of miR-122-5p mimic oligo spike-ins in assay buffer without the need for pre-amplification or pre-labelling of target nucleic acids. In this study, different chemical solutions to improve the LoD of the novel single-plex assay.⁴³ There were studied different lengths of the PEG spacer on the SMART-Base which showed no improvement in terms of LoD. On the other hand, the introduction of anionic groups at the backbone of the DGL-Probes improved hybridisation efficiencies when coupled to microspheres. By synergistically combining these technologies, the novel platform provides an accurate tool to profile miR-122-5p and enables direct detection of miRs mimicking oligo spike-ins in assay buffer in a single device.⁴³

In 2020, the merging of the DCL with Luminex xMAP technology was optimized for single-plex assay of miR-122-5p in clinical samples by introducing the unique lysis buffer called 'Stabiltech' which enabled the liberation and stabilization of miR-122-5p.⁵¹ The analytical sensitivity of the assay was determined by creating a calibration curve using known quantities of mimic miR-122-5p oligomer spike-ins in the serum matrix. No spiked serum matrix was added as the negative control and experiments were performed in triplicate.⁵¹ The MFI values obtained for the five quantities of spike-ins were, respectively, 2763.3 ± 180.3 FI, 1154.8 ± 38.5 FI, 449.8 ± 16.5 FI, 181.0 ± 6.1 FI, 86.7 ± 2.5 , and 32.0 ± 2.0 for the negative control. The curve was obtained by plotting the MFI average values (response) *versus* the quantity of spike-in(s) in fmoles (see Fig. 5).

In this study, the "Stabiltech" buffer was introduced, and additional optimisations were performed to improve the LoD to 11.1 amoles. This sensitivity level was sufficient to accurately measure miR-122-5p levels in clinical samples obtained from patients with liver injury. The optimized assay was tested on serum samples collected from patients with DILI.³⁶ The results showed that miR-122-5p was detected in all patient samples, with concentrations ranging from 1.3 to 130.3 fmoles (represented as red circles in Fig. 5), while it was not detected in healthy volunteers (not shown in Fig. 5). The study demonstrated that the assay is a reliable and feasible method for diagnosing DILI by directly quantifying miR-122-5p levels, as compared to the ALT test (data not shown).

Testing miR-122-5p, miR-451a-5p and miR-193a-5p in multi-plex

Multi-plex assays have the potential to offer a more comprehensive understanding of miR expression patterns, thereby facilitating more accurate disease diagnosis and classification, including in the case of DILI. In 2022, a study aimed at developing a method for the multi-plex detection of three miRs in serum samples using the combination of the DCL and Luminex xMAP technologies was conducted.⁵⁴ In particular, it emphasised the detection of miR-122-5p, a well-established biomarker with

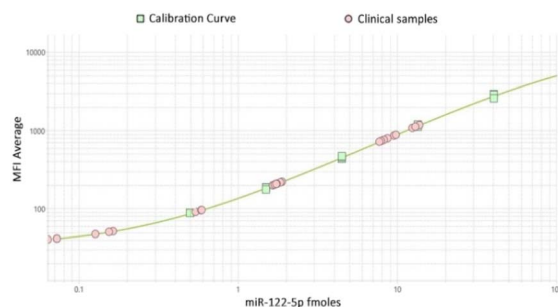


Fig. 5 Calibration curve. Using a 5PL non-linear regression model, the MFI average values were plotted vs. the quantity of synthetic miR-122-5p oligomer fmoles. Error bars (± 1 s.d.) based on triplicate measurements. The error bars are smaller than the size of some data points. $n = 3$. Calibration points generated using known quantities of mimic miR-122-5p oligomer spike-ins are represented in green squares and miR-122-5p levels in clinical samples are represented in red circles.

significant relevance in DILI diagnosis, as previously discussed. It also included miR-451a-5p, an erythroid cell-specific miR that has been linked to erythroid maturation and has shown potential as a biomarker for cancer diagnosis, prognosis, and treatment in several studies.^{73,74} Finally, miR-193a-5p was included, a miR that has been shown to be dysregulated in tumor cells and has a dual role in carcinogenesis, being up-regulated in prostate cancer but down-regulated in colorectal cancer.^{75–77} Additionally, recent studies have reported that miR-193a-5p levels increase in the serum of patients with liver injury.⁷⁸

Notably, this study demonstrated the potential of this method to simultaneously detect multiple miRs from small volumes of serum samples, which could greatly enhance our ability to diagnose and classify diseases.⁵⁴

Multi-plexing can be achieved by coupling various target-specific DGL-Probes onto Luminex microspheres with different spectral signatures. Hence, the novel platform provides a consistent, rapid, cost-effective, and straightforward method for screening multi miRs.

The study successfully demonstrated the effective coupling of DGL-Probes with Luminex microspheres and DCL reactions in both scenarios, enabling the individual interrogation of miRs in a single-plex format or simultaneous interrogation of multiple miRs in a multi-plex format. This achievement highlights the successful adaptation of DCL reagents to the Luminex xMAP platform. To validate the performance of the DGL-Probes conjugated with Luminex microspheres, DNA oligomers mimicking the three specific miRs (miR-122-5p, miR-451a-5p, and miR-193a-5p) were introduced into serum samples, and the resulting mean fluorescence intensity (MFI) values were compared between single-plex and multi-plex formats. The results revealed no significant differences in signal intensity, indicating the high compatibility of the DCL reagents with the Luminex xMAP platform for the simultaneous analysis of the three miRs together (refer to Table 2). The study effectively demonstrated that the multi-plex assay maintains the same level of sensitivity and accuracy as the single-plex assay.

After successfully developing the multi-plex method using DNA oligomers to mimic miRs, it was carried out the validation using various biological samples. The samples included serum samples from patients with DILI, serum from haemolysed blood (HS), and three pools of serums (PS1–PS3) obtained by combining samples from DILI patients and HS. The multi-plex assay incorporated the Stabiltech buffer, which was specifically designed by our group as described in ref. 51 to effectively lyse

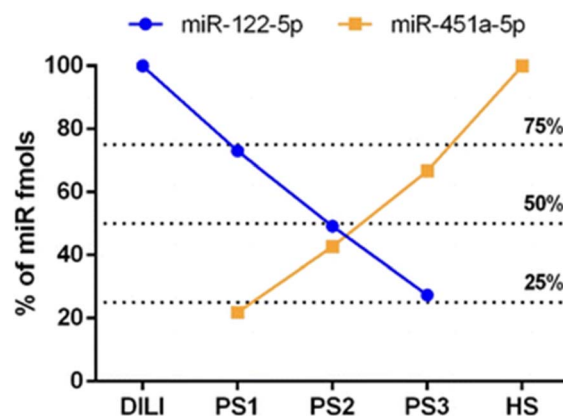


Fig. 6 Serum sample analysis ($n = 2$). miR-122-5p and miR-451a-5p detection in five types of serums: (a) DILI – a serum from a DILI patient rich in miR-122-5p; (b) HS – a serum from haemolysed blood rich in miR-451a-5p; (c) PS 1 – a pool sample generated by the mixture of a + b with a ratio of 75%/25%; (d) PS 2 – a pool sample generated by the mixture of a + b with a ratio of 50%/50%; (e) PS 3 – a pool sample generated by the mixture of a + b with a ratio of 25%/75%. y-Axis: % of miR is calculated as described in the main text of (51). Experiments were performed in duplicate for each sample. The error bars are smaller than the size of the data points and therefore not represented.

the serum samples and facilitate the release of miRs from protein complexes and extracellular vesicles.

It used Mean Fluorescence Intensity (MFI) values obtained from the samples (DILI, HS and PS1–PS3), as described in ref. 54 to perform an absolute quantification of miR-122-5p, miR-451a-5p, and miR-193a-5p by extrapolating fmoles values from a calibration curve, also described in ref. 54. The study successfully detected and quantified miR-122-5p and miR-451a-5p in the DILI, HS and PS1–PS3 samples. However, miR-193a-5p was not detected, likely because its concentration fell below the LoD of the assay, as mentioned in ref. 54. As shown in Fig. 6, the levels of miRs within DILI, HS and PS1–PS3 samples, are presented as quantities in fmols. The results revealed the following: (a) only miR-122-5p was detected in the serum of the DILI patients (with undetectable levels of miR-451a-5p); (b) only miR-451a-5p was detected in the HS (with undetectable levels of miR-122-5p); (c) none of the samples exhibited detectable levels of miR-193a-5p, indicating its absence or concentration below the assay's LoD. PS1–PS3 represents the levels of miR-122-5p *versus* miR-451a-5p, which are consistent with the volume ratios described in the legend of Fig. 6. The levels of miR-122-5p compared to miR-451a-5p in the DILI patient serums, HS, and PS1–PS3 were in accordance with the volume ratios.

Table 2 Comparison of assay sensitivity and precision between single-plex and multi-plex formats⁵⁴

miR ID	Single-plex assay		Multi-plex assay	
	Sensitivity (fmoles)	Precision	Sensitivity (fmoles)	Precision
miR-122-5p	0.05–0.20	<5%	0.05–0.20	<5%
miR-451a-5p	0.05–0.20	<5%	0.05–0.20	<5%
miR-193a-5p	Below 0.05	<5%	Below 0.05	<5%

Simultaneous measurement of arginase 1 and miR-122-5p (seqCOMBO)

An optimal biomarker combination of miRs and proteins could accurately identify DILI, predict the progression of liver injury, and help determine causality, especially for patients using multiple potentially hepatotoxic drugs. However, current technologies require splitting clinical samples into two and using different methods to test for proteins and miRs. Specifically, the

Table 3 Inter-CV comparative analysis for single-plex vs. seqCOMBO

	Single-plex ARG1 MFI average	Single-plex miR-122-5p MFI average	SeqCOMBO ARG1 MFI average	SeqCOMBO miR-122-5p MFI average	Average	SD	CV (%)
ARG1	3340.25	—	3256.0	—	3298.0	59.8	1.81
miR-122-5p	—	1376.8	—	1436.5	1406.3	42.2	3.00

first split is used for protein testing *via* an antibody-dependent or immunological assay, which takes approximately two hours, while the second split is used for miR testing *via* RT-qPCR, which involves extraction, enrichment of small RNAs, reverse transcription, and real-time amplification, taking approximately five to six hours.

In 2020, Wang *et al.* reported a new method for detecting protein and miR spike-ins, but to the best of our knowledge, simultaneous detection of both molecules from clinical samples has not been reported.³⁹ This is crucial as endogenous miRs are found in circulation within vesicles or bound to argonaute (AGO) proteins, while spike-in miRs are free molecules. Therefore, the protocol required to simultaneously detect natural miRs and proteins is different from that needed to detect spike-in miRs and proteins.

The progress achieved through the combination of DCL technology with xMAP technology has enabled the direct detection of nucleic acids, allowing for the simultaneous detection of proteins and miRs in clinical samples. By streamlining the workflow and eliminating the need for extraction, isolation, and amplification steps, DCL facilitates the detection of miRs in an enzyme-linked immunosorbent assay (ELISA) format, while preserving protein co-analytes. This breakthrough overcomes existing limitations that impede the development of highly specific and accurate simultaneous detection of proteins and miRs.

In 2021, the DCL technology was integrated with the Luminex xMAP platform to create seqCOMBO, a novel method capable of simultaneously detecting both DILI-related protein and miR molecules with high sensitivity and specificity.⁵² To demonstrate its potential utility, the seqCOMBO was applied to measure the levels of liver-type arginase 1 (ARG1) and miR-122-5p in a serum sample from a patient with DILI, using serum samples from patients without DILI as controls. As shown in Fig. 1C, the seqCOMBO workflow involved sequential interrogation of analytes, beginning with the capture of the protein biomarker, followed by the centrifugation and reservation of the pellet containing the protein, and the treatment of the remaining supernatant with Stabiltech buffer to release miR. After capturing the released miR, protein and miR microspheres were mixed again to complete the process and obtain the results.⁵²

The present study demonstrated that there were no significant signal differences between the MFI values of the single-plex and seqCOMBO methods (see Table 3), indicating the high compatibility of the antibody-dependent approach with DCL reagents on the Luminex system. Embedded in its combined technologies, seqCOMBO represents a ground-breaking diagnostic approach that can use the same patient sample to

simultaneously analyse protein and nucleic acid biomarkers of clinical significance. Despite its initial focus on DILI diagnostics, this method is poised to find new diagnostic applications beyond DILI, such as in viral diseases, where simultaneous detection of proteins and nucleic acids could provide assays with high specificity and sensitivity that exceed current capabilities.

Conclusions

This review provides valuable insights for researchers in the field of DILI and associated technologies for testing DILI-associated biomarkers. It reviews the development of DCL on Luminex xMAP technology over the years for PCR-free detection of miR-122-5p, both alone and in combination with miR-451a-5p and miR-193a-5p. Additionally, a novel method called seqCOMBO is described, which enables the simultaneous detection of circulating miR-122-5p and the ARG1 biomarker.

The integration of DCL with Luminex xMAP technology has resulted in a novel platform for rapid and accurate profiling of miR-122-5p. This platform initially demonstrated a LoD of 15 fmoles and the ability to differentiate sequences with a single nucleobase mismatch. Using the “Stabiltech” buffer and further optimisations, the assay's sensitivity improved to 11.1 fmoles, enabling direct detection and quantification of miR-122-5p levels in clinical samples from DILI patients. This assay proves to be a reliable and feasible method for diagnosing DILI, outperforming the ALT test and shows potential for the rapid and accurate diagnosis of other diseases where miRs play a critical role. However, it should be noted that the assay's sensitivity is not yet sufficient for reading the basal levels of miR-122-5p in HV.

The development of a novel method for multiplexed detection of miRs using the combination of DCL and Luminex xMAP technologies has demonstrated the potential for a more comprehensive understanding of miR expression patterns, which could lead to more accurate disease diagnosis and classification, including DILI. The multi-plex assay has shown comparable sensitivity and accuracy to the single-plex assay, making it a consistent, rapid, cost-effective, and straightforward method for clinical diagnostics and drug development through miR screening. This method enhances the ability to diagnose DILI by simultaneously detecting multiple miRs from small volumes of serum samples. While the multi-plex assay successfully detected miR-122-5p and miR-451a-5p in clinical samples, it had limitations in detecting miR-193a-5p, possibly due to its LoD. Strategies to improve sensitivity, such as exploring streptavidin conjugated with multi-phycoerythrin dyes or using a Hybridisation Chain Reaction system for amplification of the readout, are under investigation.

Optimal biomarker combinations of miRs and proteins hold great promise for accurately identifying DILI, predicting the progression of liver injury, and determining causality. The seqCOMBO method has made significant progress in this regard, enabling the direct detection of nucleic acids and facilitating the simultaneous detection of proteins and miRs in clinical samples. It shows high sensitivity and specificity in detecting DILI-related protein and miR molecules simultaneously. The compatibility of antibody-dependent approaches with DCL reagents on the Luminex system indicates the feasibility of using this method for clinical applications.

In conclusion, the combination of DCL and Luminex xMAP technologies hold great potential for assaying DILI patients through the analysis of miR-122-5p. It also enables multiplex analysis of miRs in DILI and the analysis of miRs in combination with protein biomarkers. These tools have the potential to revolutionize diagnostic approaches for DILI. Furthermore, the seqCOMBO may offer diagnostic solutions for other pathologies, including cancer and infectious diseases, where the combination of proteins and miRs can provide added value.

Author contributions

A. M. and S. P. wrote the manuscript.

Conflicts of interest

S. P. serves as a shareholder and the operations director at DESTINA Genomica S.L., a company based in Spain. Both A. M. R. and S. P. are employed by DESTINA Genomica S.L. Over the years, the company has developed the Dynamic Chemical Labelling (DCL) method for the direct analysis of single-stranded nucleic acids (ssNAs) in body fluids. DCL employs a combination of reagents: (a) biotinylated aldehyde-modified nucleobases (SMART-Bases); and (b) modified PNA capture probes synthesized with an abasic position (DGL-Probes). Both of these reagents are commercially available through DESTINA Genomica S.L.

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